CIVE 557 - Course Project

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Behaviour and Design of Steel Structures
Fall Term-2025

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1.Project Description

This report presents the structural design and detailed verification of all members and critical connections for a specified 30m span parallel chord steel truss. The work is executed in the context of a structural engineering project (CIVE 557) and adheres strictly to the requirements and provisions of the Canadian Steel Design Standard, CSA S16-19.

1.1 Project Background and Objectives

The structure analysed is truss with a total span of 30m and a constant depth of 2.5m. The truss is subjected to a defined system of factored nodal loads, including two 30 kN loads and eight 60 kN loads. As showed in figure 1,

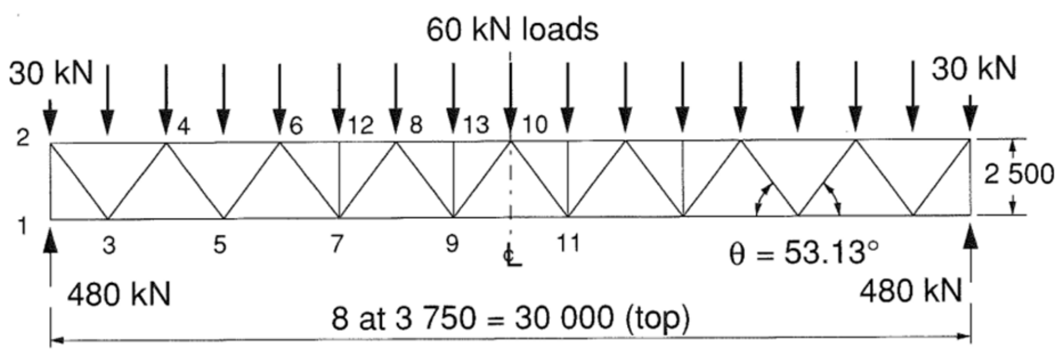

Fig.1 Truss & Load configuration

The primary objectives of this design verification are to:

1. Determine the critical maximum factored axial forces in all chord and web members based on structural analysis software SAP 2000.
2. Select appropriate **Hollow Structural Section (HSS)** member sizes that satisfy the required factored compressive Cr and tensile Tr resistances.
3. Verify the required strengths of the welded HSS-to-HSS connections for all critical nodes (i.e., connections #1 through 13), ensuring the chosen sections are adequate for both member forces and connection limit states.
4. Present a professional design report detailing assumptions, detailed calculations, and summarizing tables.

2. DESIGN CRITERIA AND ASSUMPTIONS

- This section defines the governing code, material properties, loading conditions, and fundamental structural assumptions used throughout the design and verification process.
- All member and connection strength checks are performed in accordance with the limit states design philosophy specified in the **Canadian Steel Design Standard, CSA S16-19**.

Parameter	Symbol	Value	Notes
Governing Code	N/A	CSA S16-19	Standard for the Design of Steel Structures.
Steel Grade	G40.21-M	350W	Standard structural steel grade for HSS sections.
Yield Strength (HSS)	Fy	350 MPa	Assumed for all HSS members.
Ultimate Tensile Strength (HSS)	Fu	450 MPa	Assumed for all HSS members.
Modulus of Elasticity	E	200,000 MPa	Standard value from code symbols.
Resistance Factor	**Φ**	0.90	Applied to all member and connection resistances (Clause 13.1).

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3. STRUCTURAL ANALYSIS

- 4.1 Analytical Model Setup

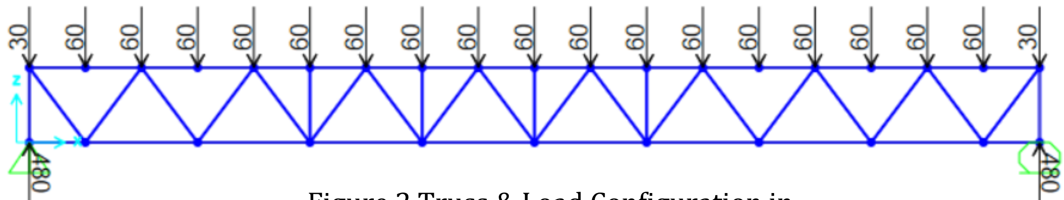


Figure 2 Truss & Load Configuration in

Now we need to input this given load combination in SAP 2000 and we are getting

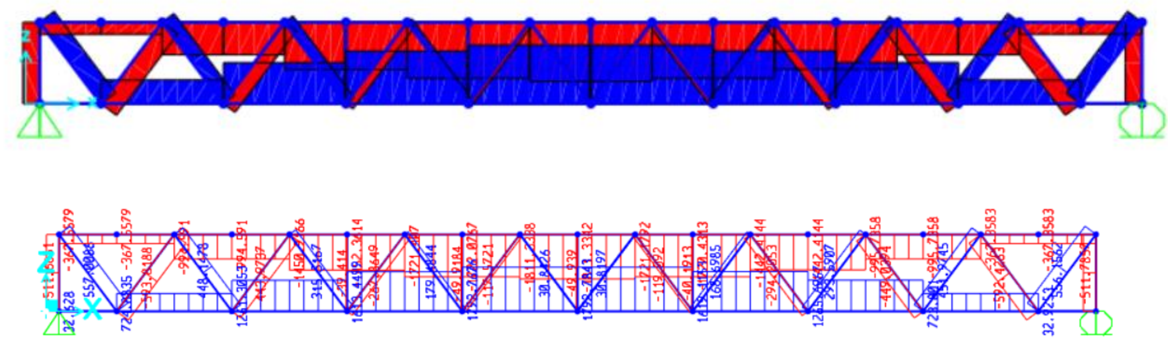


Figure 3 & 4. SAP2000 output showing axial force diagram for the truss system under applied load combinations.

The analysis indicates that the **top chord** is governed by a peak **compressive force of 1811.33 kN**, occurring symmetrically near mid-span where the truss experiences its highest flexural effects. The **bottom chord** develops a maximum **tensile force of 1792 kN** at the same location, consistent with the expected tension zone under gravity loading.

For the web members (truss branches), the critical factored axial forces are summarized as follows:

- **Maximum Compressive Force:** 593.8 kN
- **Maximum Tensile Force:** 566.13 kN

The **593.8 kN compressive force** represents the controlling demand for the design of the selected HSS web member and is used directly in evaluating member compressive resistance, slenderness limitations, and overall buckling capacity in accordance with the applicable design standard.

Member Group	Maximum Compressive Force (P_f)	Maximum Tensile Force (T_f)	Governing Design Load
Top Chord	1,811 kN	N/A (Primarily Comp.)	1,811 kN(Compression)
Bottom Chord	N/A (Primarily Tens.)	1,792 kN	1,792 kN (Tension)
Branches (Webs)	593.8 kN	566 kN	593.8 kN (Compression)

4. MEMBER DESIGN

The design was conducted referencing the CSA G40.20/G40.21 standard. Initial section selection satisfied axial load requirements but failed the critical connection checks. Final member sizes were dictated by the HSS-to-HSS connection capacity.

- Member size for top and bottom chords – 203 x 203 x 12.7 mm /Utilization Ratio – 0.6
- Member size for verticals and branches– 152 x 152 x 6.4 mm/ Utilization Ratio – 0.506

For Top and bottom chords and secondary members **Clause 13.3** and **13.2** have been considered.

5. CONNECTION DESIGN

The connection design done by using HSS Connex Software form Node #1 to 13. Also, for the eccentricity calculation Appendix 1 used.

Sr No.	Nodes	Connection Type	Utilization ratio
1	Node 1	T- Connection	0.75
2	Node 2	K- Connection	0.576
3	Node 3	K- Connection	0.559
4	Node 4	K + X + Y - Connection	0.577
5	Node 5	K- Connection	0.422
6	Node 6	K + X + Y - Connection	0.438
7	Node 7	K + T- Connection	0.355
8	Node 8	K + X + Y - Connection	0.263
9	Node 9	K + Y + T Connection	0.249

10	Node 10	K + Y - Connection	0.122
11	Node 11	K + T - Connection	0.103
12	Node 12	X- Connection	0.058
13	Node 13	X- Connection	0.074

- The HSS-to-HSS welded connections were designed as **Gapped K-Connections**. Based on the required wall thickness and geometric constraints of the AISC/CSA code, an **Eccentricity of 43 mm** was strictly maintained at all internal nodes (Nodes # 3 through 11). This minimum gap value governs the joint detailing and is necessary to satisfy the **limits of applicability** for the Chord Surface Plastification yield line equations. Any reduction in this gap would result in the connection failing the geometric assumptions of the design code

6. Comments and Interpretation

- Review of the connection utilization ratios confirms that all joints operate within permissible limits, with the maximum recorded ratio (0.75 at Node 1) remaining below the φ-factored resistance threshold.
- HSS Connex output validates that the applied gap and overlap configurations satisfy the geometric requirements for HSS K, T, and Y connections, and the chord face plastification, punching shear, and brace-to-chord checks remain within acceptable ranges.
- Member axial demands obtained from the global analysis are well within the available capacities per CSA S16/ AISC, indicating no exceedance of factored compression or tension limits.
- Overall structural adequacy is confirmed, with both member-level and connection-level capacity checks meeting strength and serviceability requirements for the governing load combinations.

7. REFERENCES

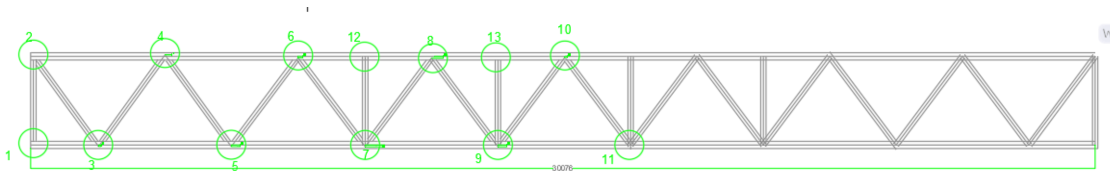
- CSA S16-19 – Design of Steel Structures, Canadian Standards Association, 2019.
- CSA G40.20/G40.21-19 – General Requirements for Rolled or Welded Structural Quality Steel, Canadian Standards Association, 2019.
- AISC Steel Desing Guide 24 – Packer, J. A., and E. W. H. Ma. Design Guide 24: Hollow Structural Section Connections. Chicago, IL: AISC, 2010.

8.Software Utilization

- SAP2000 v24, Computers and Structures, Inc., 202.
- HSS Connex v6.3, American Institute of Steel Construction, 2023
- AutoCAD final technical drafting and detailing of the members and joints for fabrication.

9.APPENDICES

A. Auto-Cad Draft for Eccentricity calculation



Typical Truss

Truss Member Details			
Sl. No	Section	Member	Unit Wt. (Kg/m)
1	203 x 203 x 12.7	Top & Bot. Chord	56
2	152 x 152 x 6.4	Branches	28
3	152 x 152 x 6.4	Verticals	28

Figure A.1.AutoCad drafting for an Eccentricity correction

B. HSS Connex Sort ware Output for Node #1 to 13

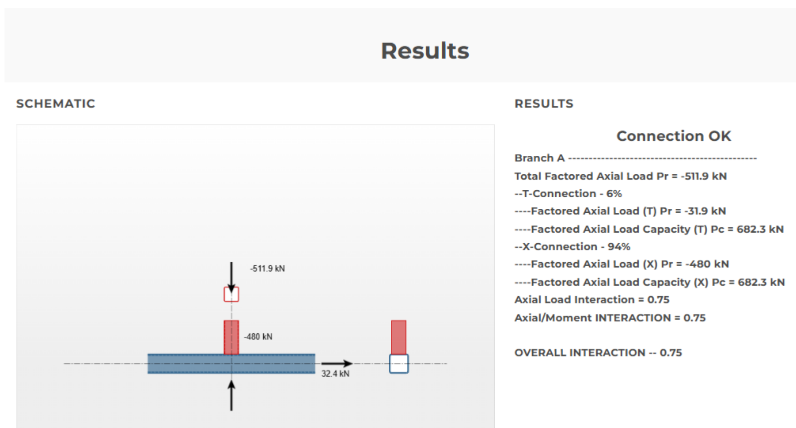


Figure B.1. HSS Connex output at Node #1

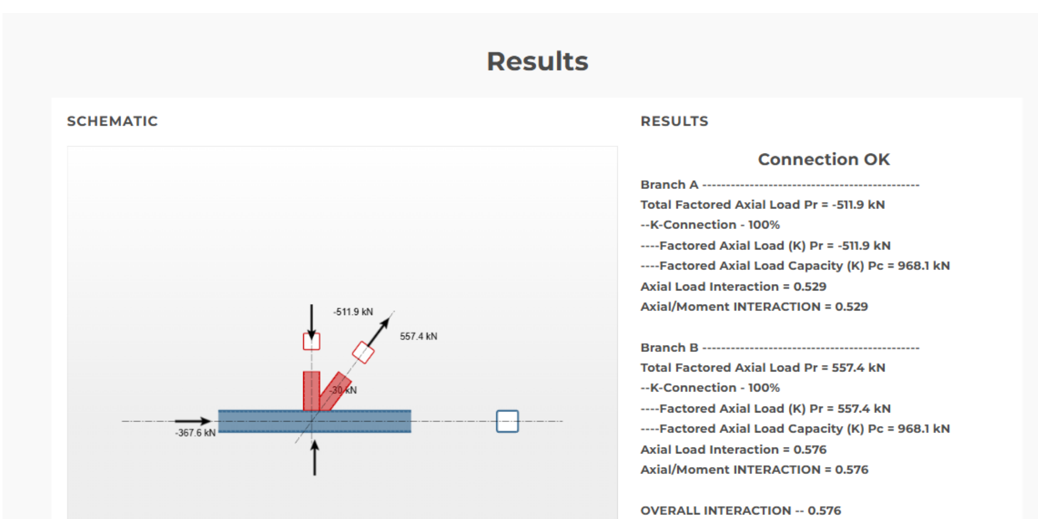


Figure B.2. HSS Connex output at Node #2

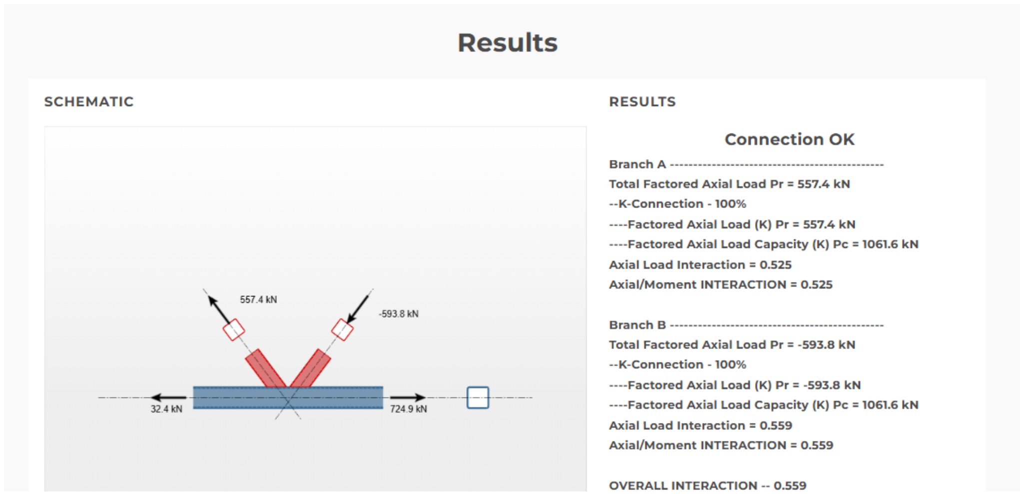


Figure B.3. HSS Connex output at Node #3

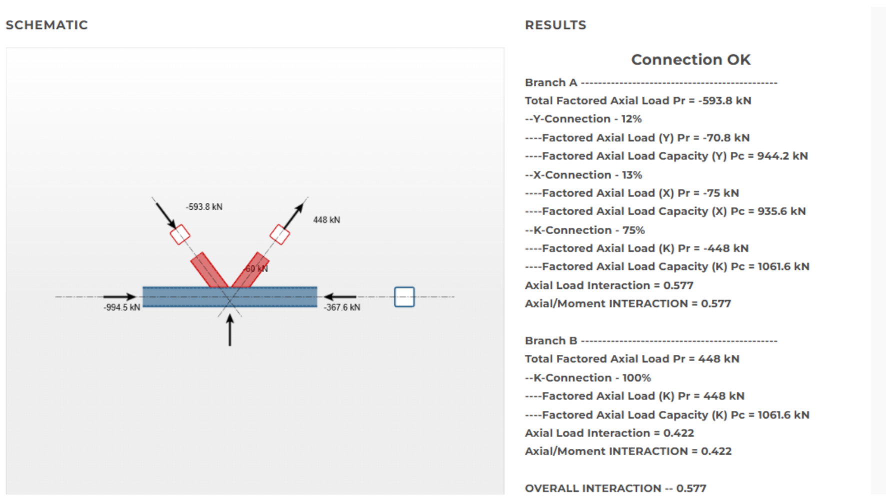


Figure B.4. HSS Connex output at Node #4

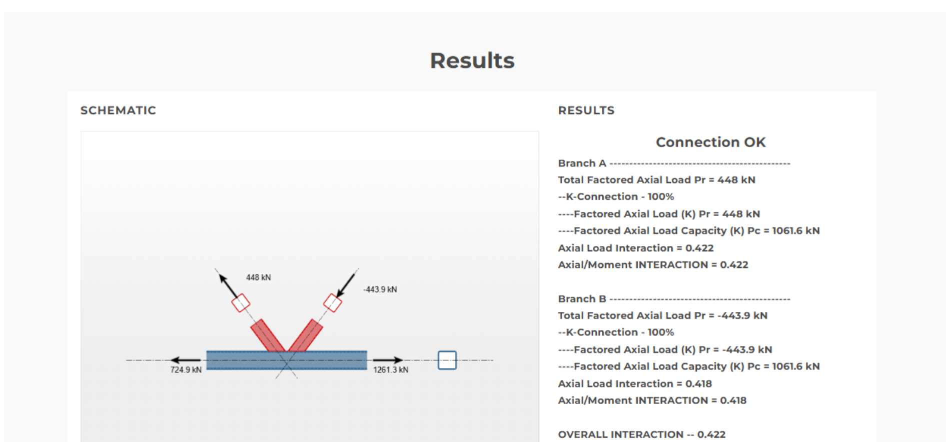


Figure B.5. HSS Connex output at Node #5

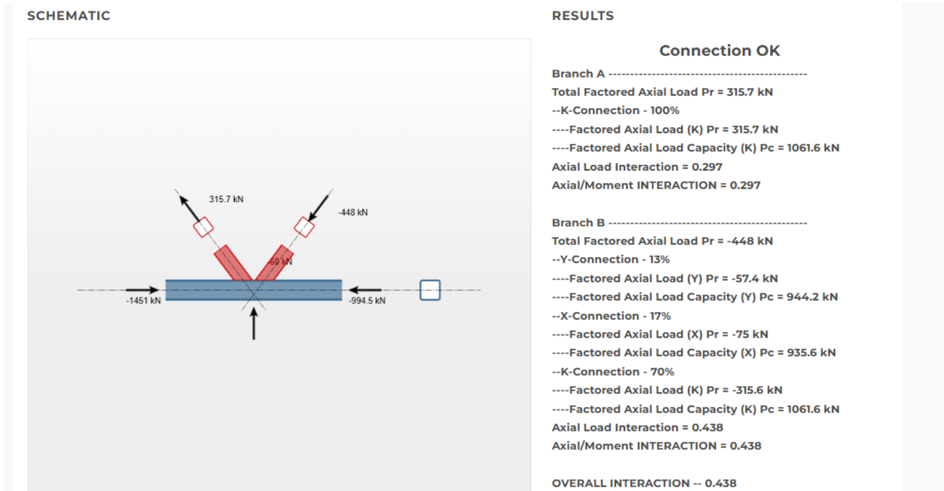


Figure B.6. HSS Connex output at Node #6

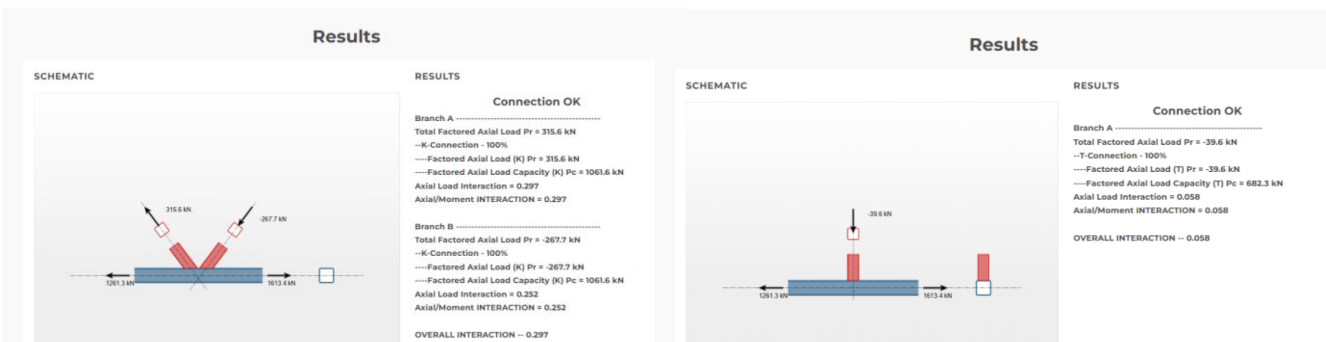


Figure B.7. HSS Connex output at Node #7

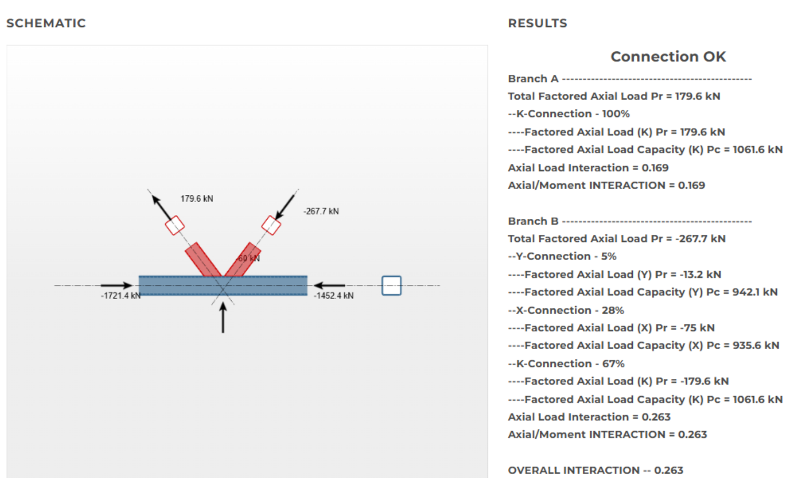
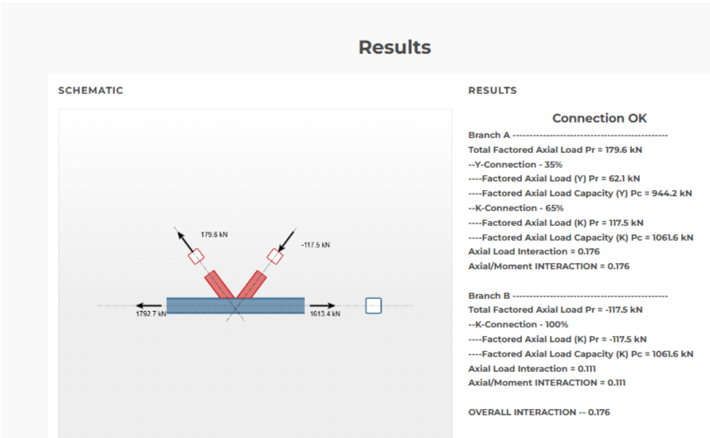


Figure B.8. HSS Connex output at Node #8



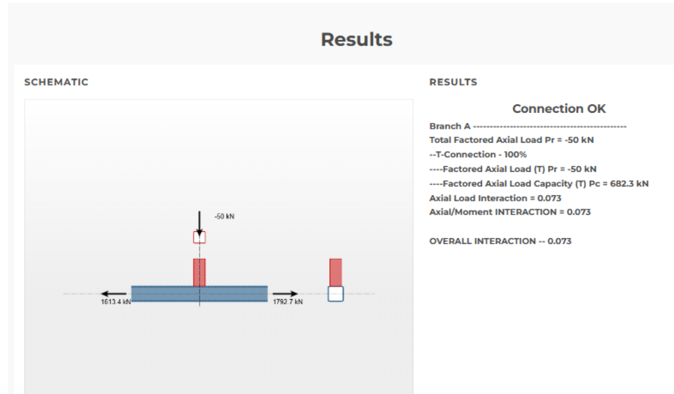


Figure B.9. HSS Connex output at Node #9

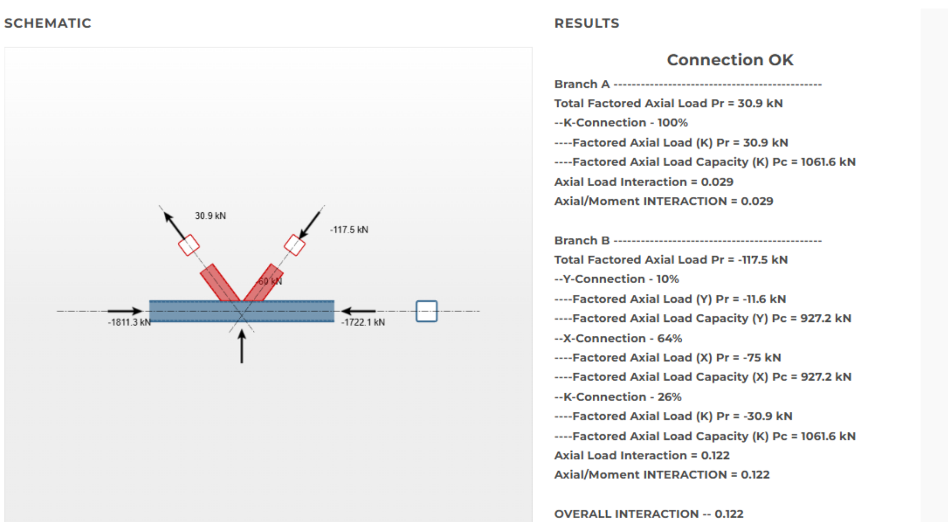
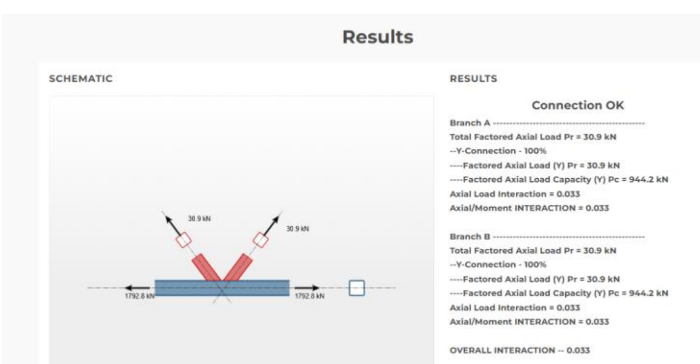


Figure B.10. HSS Connex output at Node #10



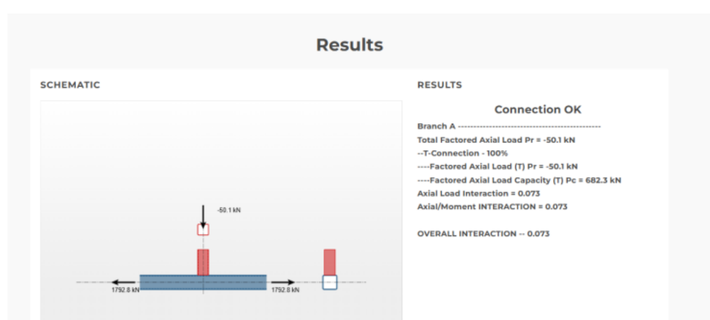


Figure B.11. HSS Connex output at Node #11

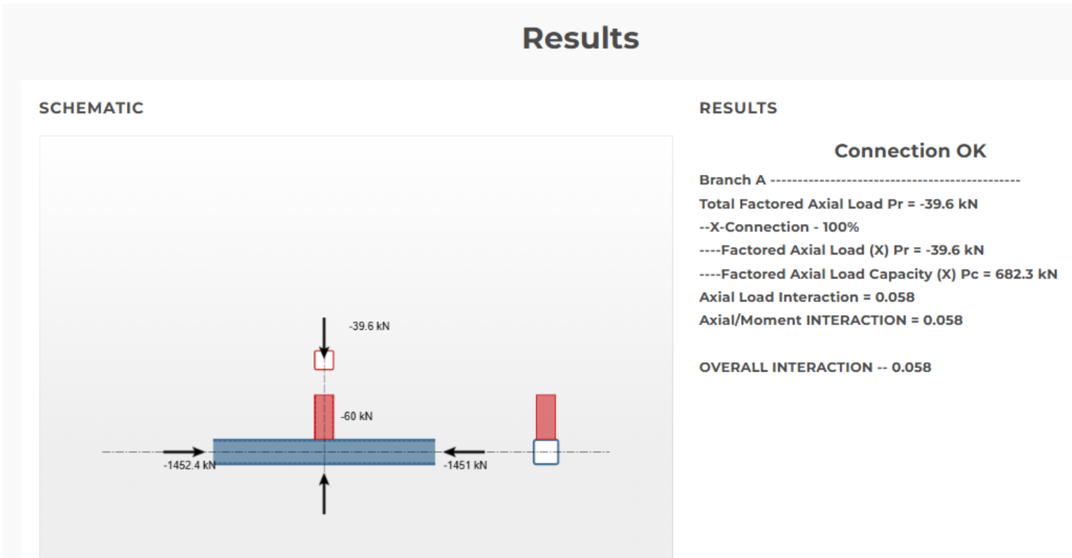


Figure B.12. HSS Connex output at Node #12

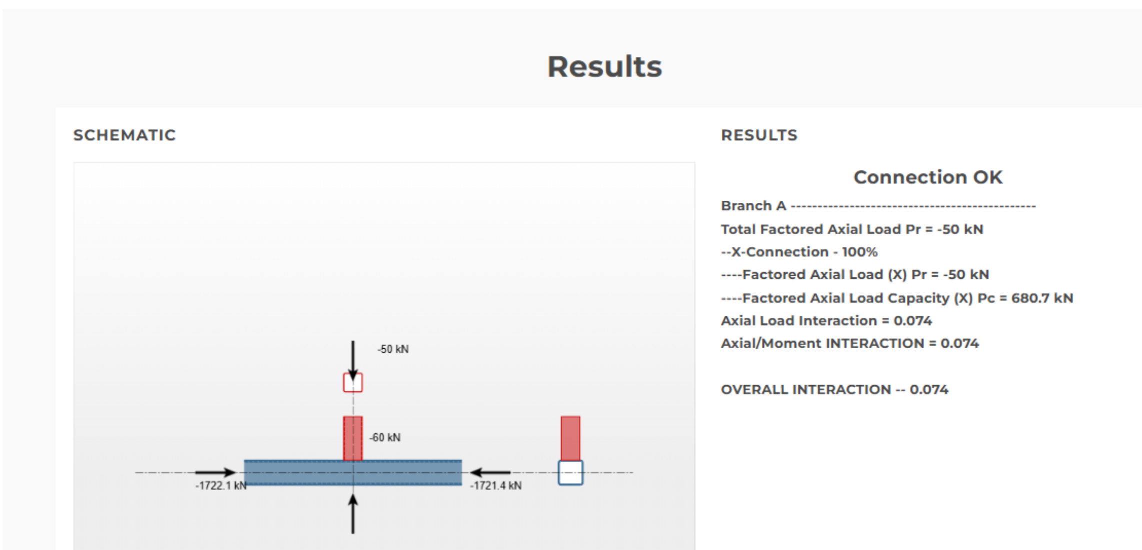


Figure B.13. HSS Connex output at Node #13